

# Aleksei Gregory Sorokin

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**Research** Scientific Machine Learning, Gaussian Processes, Quasi-Monte Carlo, Probabilistic Numerics

**Programming** Python (PyTorch, GPyTorch, Pandas, Matplotlib), Julia, C, MATLAB, R, SQL, Wolfram

**Tools** AWS (SageMaker, EC2), GitHub (Actions, Pages), L<sup>A</sup>T<sub>E</sub>X, Docker

## Education

2026.01 – 2028.05 **Postdoc in Statistics**. University of Chicago (UChicago). Advisors *Yuehaw Khoo and Lek-Heng Lim*.

2021.08 – 2025.12 **PhD in Applied Math**. Illinois Institute of Technology (IIT). GPA 3.89/4. Advisor *Fred J. Hickernell*.

2017.08 – 2021.05 **Master of Data Science**. IIT. Summa Cum Laude. GPA 3.94/4.

2017.08 – 2021.05 **B.S. in Applied Math, Minor in Computer Science**. IIT. Summa Cum Laude. GPA 3.94/4.

## Experiences

2025.01 – 2025.12 **DOE SCGSR Fellow in Applied Math at Sandia National Laboratories**, Livermore, CA. I produced scientific ML models for machine-precision solutions to nonlinear PDEs [1]. I developed scalable multifidelity Gaussian process regression models and open-source software implementations [10, 11, 7].

2021.08 – 2025.01 **Teaching Assistant at IIT**. I led reviews for PhD qualifying exams in analysis and computational math.

2024.05 – 2024.08 **Scientific Machine Learning Researcher at FM (Factory Mutual Insurance Company)**. I deployed scientific ML models, including PINNs and DeepONets, to accelerate CFD fire simulations [13].

2023.05 – 2023.08 **Graduate Intern at Los Alamos National Laboratory**. I modeled multifidelity solutions to PDEs with random coefficients using efficient and error-aware Gaussian process regression models [14].

2022.09 – 2022.11 **Participant in Argonne National Laboratory's Course on AI-Driven Science on Supercomputers**. Key topics included handling large-scale data pipelines and parallel training for neural networks.

2022.05 – 2022.08 **Givens Associate Intern at Argonne National Laboratory**. I derived error bounds and a sequential sampling method for efficiently estimating failure probabilities with probabilistic models [6].

2021.05 – 2021.08 **ML Engineer Intern at SigOpt (acquired by Intel)**. On a six-person ML team, I contributed production code for meta-learning model-aware hyperparameter tuning via Bayesian optimization [18].

2018.05 – 2019.08 **Instructor of the Computer Discovery Program, STARS Computing Corps at IIT**. I taught and developed curriculum for middle school and high school girls to learn programmatic thinking in Python.

## Open-Source Software

**QMCPy Quasi-Monte Carlo Python Software** (qmcsoftware.org/qmcpy). I led dozens of collaborators across academia and industry to develop QMC sequence generators, automatic variable transformations, adaptive error estimators, and diverse use cases [5, 8, 9, 12, 19, 15, 20, 3, 4].

**FastGPs Scalable Gaussian Processes in Python** (alegresor.github.io/fastgps). FastGPs pairs lattice/digital-net designs with matching shift-invariant kernels, reducing GP costs from  $\mathcal{O}(n^3)$  time and  $\mathcal{O}(n^2)$  storage to  $\mathcal{O}(n \log n)$  time and  $\mathcal{O}(n)$  storage. This supports GPU scaling, batched inference, hyperparameter optimization, multifidelity GPs, and efficient Bayesian cubature [11, 10, 7].

**TorchOrthoPolys Orthogonal Polynomials in PyTorch** (alegresor.github.io/TorchOrthoPolys) with GPU support.

**QMCToolsCL Randomized Quasi-Monte Carlo Sequences in C/OpenCL** (qmcsoftware.org/QMCToolsCL).

**QMCGenerators.jl Randomized Quasi-Monte Carlo Sequences in Julia** (alegresor.github.io/QMCGenerators.jl).

## Awards

2025.01 – 2025.12 **DOE SCGSR Fellow in Applied Math**, Sandia National Laboratories, Livermore, CA.

2025.01 **Karl Menger Student Award for Exceptional Scholarship (Graduate)**, IIT.

2024.01 **College of Computing Excellence in Dissertation Research**, IIT.

2024 **Teaching Assistant Award**, IIT.

2023.08 **Outstanding Math Poster**, Los Alamos National Laboratory.

2021 **Best Manuscript**, IIT Undergraduate Research Journal.

2020 **Karl Menger Student Award for Exceptional Scholarship (Undergraduate)**, IIT.

2017.08 – 2025.05 **Dean's List Member**, IIT, every semester.

## References

PhD Advisor **Fred J. Hickernell** (hickernell@illinoistech.edu) Professor of Applied Math, IIT.

Fellowship Mentor **Pieterjan M. Robbe** (pmrobbe@sandia.gov) Senior Member of Technical Staff, Sandia National Labs.

Postdoc Mentor **Yuehaw Khoo** (ykhoo@uchicago.edu) Assistant Professor, Department of Statistics, UChicago.

Postdoc Mentor **Lek-Heng Lim** (lekheng@uchicago.edu) Professor, Department of Statistics, UChicago.

Collaborator **Sou-Cheng T. Choi** (schoi32@illinoistech.edu) Research Associate Professor of Applied Math, IIT.

Supervisor, Intel **Michael J. McCourt** (mikemccourt1234@gmail.com) Director of ML Ops, Torc Robotics.

## Academic Service: Peer Reviews

2026 **SIAM Book Program**, Society for Industrial and Applied Mathematics.

2026 **PMLR-ProbNum**, Proceedings of the Second International Conference on Probabilistic Numerics.

2026 **SINUM**, SIAM Journal on Numerical Analysis.

2025 **PMLR-ProbNum**, Proceedings of the First International Conference on Probabilistic Numerics.

2022 **MCQMC**, Monte Carlo and Quasi-Monte Carlo Methods 2022.

## Conference Organization

2027 **American Mathematical Society (AMS) Joint Mathematics Meetings (JMM)**. Organizer of the session on “*High-Dimensional Sampling: Numerical Integration, Dynamics, and Surrogate Modeling*”.

2027 **American Mathematical Society (AMS) Spring Central Sectional Meeting**. Organizer of the session on “*Tensors, Transport, and Particle Dynamics in High-Dimensional Sampling and Inference*”.

2025 **International Conference on Monte Carlo Methods and Applications**. Conference organization assistant and chair of the session on “*Hardware or Software for (Quasi-)Monte Carlo Algorithms*”.

2024 **International Conference on Monte Carlo and Quasi-Monte Carlo Methods in Scientific Computing**. Organizer of the session on “*Efficient Bayesian Surrogate Modeling*”.

## Coursework

**Math** Applied Analysis I/II, Computational Math, Probability, Statistics, Applied Statistics, Bayesian Computational Statistics, Statistical Learning, Monte Carlo Methods in Finance, Mathematical Methods for Algorithmic Trading, Numerical Methods for PDEs, Reliable Mathematical Software, Linear Optimization, Computational Algebraic Geometry.

**Computer Science** Big Data Technologies, Data Preparation and Analysis, Database Organization, Big Data Visualization, Systems Programming, Computer Organization and Assembly, Data Structures and Algorithms, Object-Oriented Programming I/II.

## Publications

- [1] Aras Bacho, Aleksei G. Sorokin, Xianjin Yang, Théo Bourdais, Edoardo Calvella, Matthieu Darcy, Alexander W. Hsu, Bamdad Hosseini, and Houman Owhadi. “Operator learning at machine precision”. In: *Journal of Computational Physics* 566 (2026), p. 115240. DOI: 10.1016/j.jcp.2026.115240.
- [2] Eda Gjergo, Zhi-Yu Zhang, Pavel Kroupa, Aleksei G. Sorokin, Zhiqiang Yan, Ziyi Guo, Tereza Jerabkova, Akram Hasani Zonoozi, and Hosein Haghi. “Massive Star Formation at Supersolar Metallicities: Constraints on the Initial Mass Function”. In: *Research in Astronomy and Astrophysics* 26.2 (Jan. 2026), p. 025003. DOI: 10.1088/1674-4527/ae1f79.
- [3] Fred J. Hickernell, Nathan Kirk, and Aleksei G. Sorokin. “Quasi-Monte Carlo methods: what, why, and how?” In: *Monte Carlo and Quasi-Monte Carlo Methods 2024*. Ed. by Christiane Lemieux and Ben Feng. Springer, 2026, pp. 3–40. DOI: 10.1007/978-3-032-10590-5\_1.
- [4] Aadit Jain, Fred J. Hickernell, Art B. Owen, and Aleksei G. Sorokin. “Empirical Bernstein and betting confidence intervals for randomized quasi-Monte Carlo”. In: *Information and Inference: A Journal of the IMA* 15.1 (Mar. 2026), iaag003. DOI: 10.1093/imaiai/iaag003.
- [5] Aleksei G. Sorokin, Fred J. Hickernell, Sou-Cheng T. Choi, Jagadeeswaran Rathinavel, Pieterjan Robbe, and Aadit Jain. “QMCPy: a Python framework for (quasi-)Monte Carlo algorithms”. In: *Journal of Open Source Software* 11.117 (2026), p. 9705. DOI: 10.21105/joss.09705.
- [6] Aleksei G. Sorokin and Vishwas Rao. *Credible intervals for probability of failure with Gaussian processes*. 2026. DOI: 10.48550/arXiv.2311.07733.
- [7] Aleksei G. Sorokin, Pieterjan Robbe, Yen-Chun Liu, Simon Mak, and Fred J. Hickernell. *Fast multitask Gaussian processes, with application to surrogate modeling of the quark-gluon plasma*. 2026. DOI: 10.48550/arXiv.2603.16014.
- [8] Aleksei G. Sorokin. “Algorithms and scientific software for quasi-Monte Carlo, fast Gaussian process regression, and scientific machine learning”. PhD thesis. Illinois Institute of Technology, 2025. DOI: 10.48550/arXiv.2511.21915.
- [9] Aleksei G. Sorokin. *QMCPy: a Python package for randomized low-discrepancy sequences, quasi-Monte Carlo, and fast kernel methods*. 2025. DOI: 10.48550/arXiv.2502.14256.
- [10] Aleksei G. Sorokin, Pieterjan Robbe, Gianluca Geraci, Michael S. Eldred, and Fred J. Hickernell. *Fast Bayesian multilevel quasi-Monte Carlo*. 2025. DOI: 10.48550/arXiv.2510.24604.

- [11] Aleksei G. Sorokin, Pieterjan Robbe, and Fred J. Hickernell. "Fast Gaussian process regression for high dimensional functions with derivative information". In: *Proceedings of the First International Conference on Probabilistic Numerics*. Ed. by Motonobu Kanagawa, Jon Cockayne, Alexandra Gessner, and Philipp Hennig. Vol. 271. Proceedings of Machine Learning Research. PMLR, 2025, pp. 35–49. URL: <https://proceedings.mlr.press/v271/sorokin25a.html>.
- [12] Sou-Cheng T. Choi, Yuhan Ding, Fred J. Hickernell, Jagadeeswaran Rathinavel, and Aleksei G. Sorokin. "Challenges in developing great quasi-Monte Carlo software". In: *Monte Carlo and Quasi-Monte Carlo Methods 2022*. Ed. by Aicke Hinrichs, Peter Kritzer, and Friedrich Pillichshammer. Springer, 2024, pp. 209–222. DOI: 10.1007/978-3-031-59762-6\_9.
- [13] Aleksei G. Sorokin, Xiaoyi Lu, and Yi Wang. "A neural surrogate solver for radiation transfer". In: *NeurIPS 2024 Workshop on Data-Driven and Differentiable Simulations, Surrogates, and Solvers*. 2024. URL: <https://openreview.net/forum?id=SHidR8UMKo>.
- [14] Aleksei G. Sorokin, Aleksandra Pachalieva, Daniel O'Malley, James M. Hyman, Fred J. Hickernell, and Nicolas W. Hengartner. "Computationally efficient and error aware surrogate construction for numerical solutions of subsurface flow through porous media". In: *Advances in Water Resources* 193 (2024), p. 104836. DOI: 10.1016/j.advwatres.2024.104836.
- [15] Aleksei G. Sorokin and Jagadeeswaran Rathinavel. "On bounding and approximating functions of multiple expectations using quasi-Monte Carlo". In: *Monte Carlo and Quasi-Monte Carlo Methods 2022*. Ed. by Aicke Hinrichs, Peter Kritzer, and Friedrich Pillichshammer. Springer, 2024, pp. 583–599. DOI: 10.1007/978-3-031-59762-6\_29.
- [16] Eda Gjergo, Aleksei G. Sorokin, Anthony Ruth, Emanuele Spitoni, Francesca Matteucci, Xilong Fan, Jinning Liang, Marco Limongi, Yuta Yamazaki, Motohiko Kusakabe, and Toshitaka Kajino. *GalCEM: GALactic Chemical Evolution Model*. Astrophysics Source Code Library, record ascl:2301.022. Jan. 2023. URL: <https://ui.adsabs.harvard.edu/abs/2023ascl.soft01022G>.
- [17] Eda Gjergo, Aleksei G. Sorokin, Anthony Ruth, Emanuele Spitoni, Francesca Matteucci, Xilong Fan, Jinning Liang, Marco Limongi, Yuta Yamazaki, Motohiko Kusakabe, and Toshitaka Kajino. "GalCEM. I. An Open-source Detailed Isotopic Chemical Evolution Code". In: *The Astrophysical Journal Supplement Series* 264.2 (Feb. 2023), p. 44. DOI: 10.3847/1538-4365/aca7c7.
- [18] Aleksei G. Sorokin, Xinran Zhu, Eric Hans Lee, and Bolong Cheng. "SigOpt Mulch: an intelligent system for AutoML of gradient boosted trees". In: *Knowledge-Based Systems* 273 (2023), p. 110604. DOI: 10.1016/j.knosys.2023.110604.
- [19] Sou-Cheng T. Choi, Fred J. Hickernell, Jagadeeswaran Rathinavel, Michael J. McCourt, and Aleksei G. Sorokin. "Quasi-Monte Carlo software". In: *Monte Carlo and Quasi-Monte Carlo Methods 2020*. Ed. by Alexander Keller. Springer, 2022, pp. 23–47. DOI: 10.1007/978-3-030-98319-2\_2.
- [20] Aleksei G. Sorokin, Fred J. Hickernell, Sou-Cheng T. Choi, Michael J. McCourt, and Jagadeeswaran Rathinavel. "(Quasi-)Monte Carlo Importance Sampling with QMCPy". In: *IIT Undergraduate Research Journal* (2021), pp. 49–54. URL: <http://urj.library.iit.edu/index.php/urj/article/view/48>.